Assigned: Friday Sept. 11, 2026
Due: Friday Sept. 25, 2026

MATH 6660: Homework 2
Fall 2026

CG(1) for 2-pt BVP with periodic BCs:

Consider the following boundary value problem with periodic boundary conditions:

$$\begin{aligned} \textbf{ODE}: &\quad -u'' + q(x)\,u = f(x), \\ \textbf{BC1}: &\quad u(-\pi) = u(\pi), \\ \textbf{BC2}: &\quad u'(-\pi) = u'(\pi), \end{aligned}$$

where $q > 0\ \forall x \in [-\pi, \pi]$.

1. Derive a Galerkin orthogonality expression for this problem and deduce from this an appropriate *energy norm*.

2. Develop a CG(1) method for this problem, clearly stating the appropriate spaces, trial, test, and basis functions.

3. Show that the CG(1) FEM solution gives the optimal solution in the *energy norm*. Use this to prove an *a priori* error estimate in the *energy norm*.

4. Prove an *a priori* error estimate in the L^2-norm.

5. Numerically implement the CG(1) method on a uniform mesh with the following functions

$$q(x) = 3 - \sin(x) - \sin(2x) - \cos(x), \quad f(x) = 2e^{\sin(x)}e^{\cos(x)}, \quad u(x) = e^{\sin(x)}e^{\cos(x)}.$$

   Numerically determine the convergence rate in both the *energy norm* and the L^2-norm by creating an appropriate convergence table with

$$M = 24, 49, 99, 199, 399, 799.$$

   HINT: You can assume a uniform grid spacing $h = 2\pi/(M+1)$. For a given M, you will need to write Python code that assembles the stiffness matrix, $\underline{\underline{A}}$, and the right-hand side vector, $\underline{b}$. I recommend that you store $\underline{\underline{A}}$ as a sparse array using `scipy.sparse` and that you solve the resulting linear system using the built-in conjugate gradient solver `cg` from `scipy.sparse.linalg`. You can find documentation for how to use these functions here: `https://docs.scipy.org/doc/scipy/`.

CG(2) for 2-pt BVP with Dirichlet BCs:

Consider the following boundary value problem with periodic boundary conditions:

$$\begin{aligned} \textbf{ODE}: &\quad -u'' = f(x), \\ \textbf{BC}: &\quad u(0) = u(1) = 0. \end{aligned}$$

6. Implement the CG(2) (i.e., piecewise quadratic polynomials) method as discussed in class for this problem.

   HINT: You can assume a uniform grid spacing $h = 1/(M+1)$. For a given M, you will need to write Python code that assembles the stiffness matrix, $\underline{\underline{A}}$, and the right-hand side vector, $\underline{b}$. I recommend that you store $\underline{\underline{A}}$ as a sparse array using `scipy.sparse` and that you solve the resulting linear system using the built-in conjugate gradient solver `cg` from `scipy.sparse.linalg`. You can find documentation for how to use these functions here: `https://docs.scipy.org/doc/scipy/`.

7. Apply your method to the following example:

   $$u(x) = e^{1-\cos(2\pi x)-\sin(2\pi x)} - 1, \quad f(x) = 4\pi^2 e^{1-\cos(2\pi x)-\sin(2\pi x)} \Big[\sin(4\pi x) - \cos(2\pi x) - \sin(2\pi x) - 1\Big].$$

   Numerically determine the convergence rate in both the *energy norm* and the L^2-norm by creating an appropriate convergence table with

   $$M = 24, 49, 99, 199, 399, 799.$$